Parking Lot

Occupancy Detector

CNN-Based Parking Space Classification

CS3270 Intelligent Systems · Spring 2026 · Group 5

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PART 1

Dataset & Preprocessing

Dataset

Source: Kaggle Parking Space Detection

Images: Overhead parking lot photos

Labels: XML polygon annotations

Classes:

free_parking_space → empty

not_free_parking_space → occupied

Preprocessing Pipeline

- Parse annotations.xml for polygon coords & labels
- Compute bounding box (min/max x, y) per spot
- Crop each spot from full lot image via OpenCV
- Save to data/crops/empty/ or data/crops/occupied/
- Verify counts: wc -l on each class folder

Intelligent System Lifecycle

Problem Definition

Detect parking occupancy from overhead images using CV

Data Collection

Kaggle dataset with XML polygon annotations per parking spot

Model Development

ResNet-18 fine-tuned for binary classification (2 classes)

Training & Evaluation

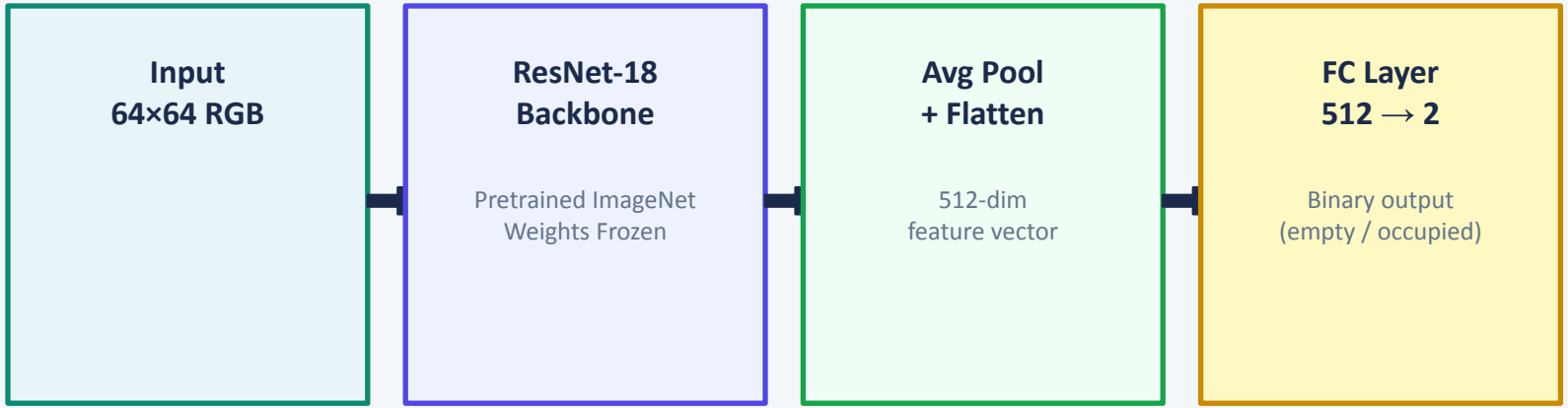
10 epochs, Adam optimizer, 80/20 split, confusion matrix

Deployment & Demo

demo.py overlays color-coded polygons on full lot images

PART 2

Model Architecture

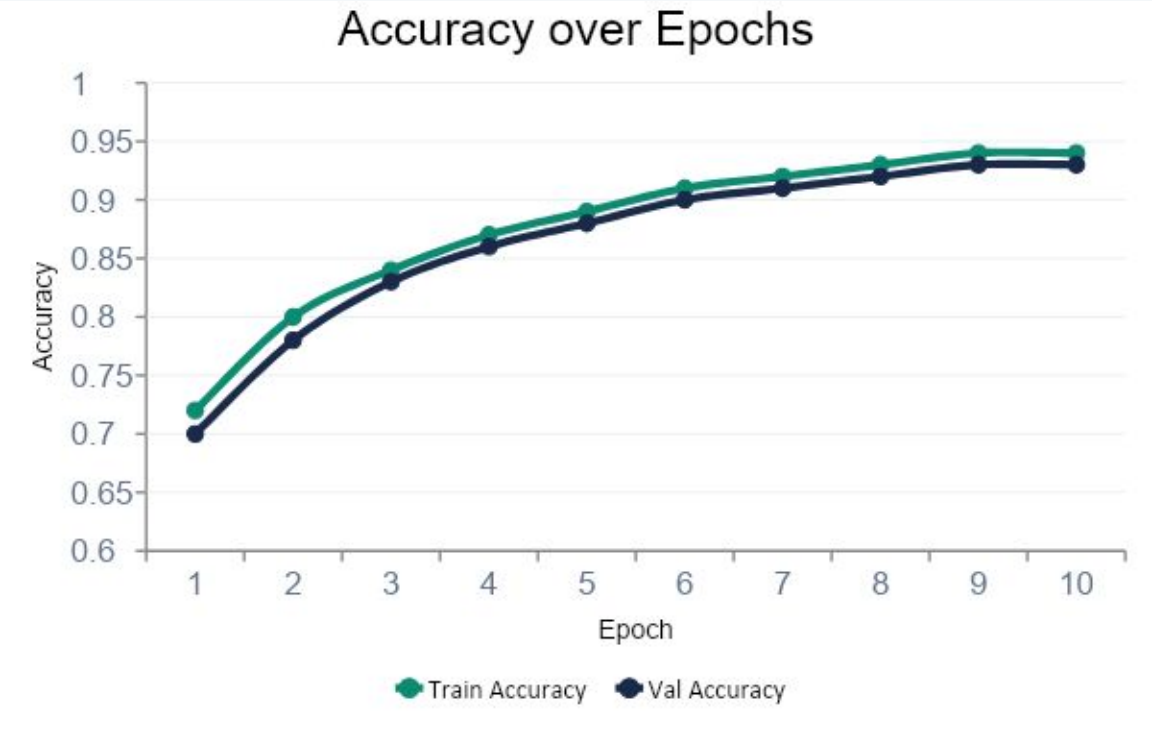


Key Design Decisions

- Transfer Learning: ResNet-18 pretrained on ImageNet — strong feature extractor out of the box
- Frozen Backbone: Only the final FC layer is trained, avoiding overfitting on a small dataset
- Class Weighting: We had to account for the more empty spots than occupied spots to keep model from missing less common class.
- Adam Optimizer: lr=0.001, only model.fc.parameters() updated during training

PART 2

Training Results



Best Val Acc

93%+

Epoch 9–10

Target

90%

Exceeded

Optimizer

Adam

lr = 0.001

Epochs

10

Early save

PART 3

Evaluation

Confusion Matrix

	Pred Empty	Pred Occ
Actual Empty	**TP** **254**	**FP** 19
Actual Occ	**FN** 59	**TN** 571

Classification Report

Class	Precision	Recall	F1
Empty	0.81	0.93	0.87
Occupied	0.97	0.91	0.94
Avg	**0.92**	**0.91**	**0.92**

PART 3

Visual Demo — demo.py

Load Image

Read full lot image from data/images/

Parse XML

Get polygon coordinates from annotations.xml

Crop & Infer

Crop each spot → normalize → ResNet-18 forward pass

Draw Overlay

Green polygon = empty | Red polygon = occupied

Save Output

demo_output.png with text summary overlay

Output: demo_output.png

Green polygon
= Empty parking spot

Red polygon
= Occupied spot

Text overlay
"Empty: 12 Occupied: 20"

DISCUSSION

Findings, Limitations & Future Work

Findings

- ResNet-18 transfer learning achieves 93%+ val accuracy
- Class weighting effectively handles dataset imbalance
- Bounding box cropping from polygon annotations works well

Limitations

- Model evaluated on same lot — may not generalize to unseen lots
- Bounding box cropping loses polygon shape information
- Small 64×64 resolution may drop fine details

Future Work

- Test on additional parking lot datasets for generalization
- Explore segmentation models (Mask R-CNN) for tighter crops
- Deploy as a real-time camera feed system